

Machine-Learning Application for Detection and Prediction of Stroke Risk and Post-Stroke Outcomes: A Comprehensive Review

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Abstract— The integration of digital technologies in healthcare has piloted in a new era of utilising machine learning models, enabling early detection and prediction of disease occurrences. Despite these advancements, efficiently predicting and detecting the risk of stroke and post-stroke outcomes remains a significant challenge. Stroke often goes undetected until after it has occurred, and its potential consequences are largely unknown. Researchers have proposed using Machine-Learning (ML) models to detect and predict stroke occurrence and post-stroke outcomes to address these challenges. To the best of our knowledge, while there has been considerable research on stroke prediction, a thorough combination of ML applications intended for both stroke prediction and post-stroke outcomes remains notably absent in the literature. This paper offers an extensive literature review of ML applications in stroke prediction, post-stroke outcomes, contributory risk factors, and detection and prediction methodologies, aiming to bridge existing gaps in knowledge within this domain. This study assessed forty-six research studies authored from 2016 to 2024, aiming on ML applications for detecting and predicting stroke risks and/or post-stroke outcomes, carefully chosen from broad range of over 800 research studies through a comprehensive analysis of both quantitative and qualitative data. By synthesising and analysing research outcomes of these studies, this study intent to deliver perspective on the present status of research in this domain, identify key trends and challenges, and offer recommendations for future research directions. This review paper aims to enrich the current body of knowledge in the domain of stroke prediction and post-stroke outcomes through the lens of ML applications. Its overarching goal is to enhance early detection and predictive methodologies, thereby lessening the impact of stroke in Africa and globally.

Keywords— Stroke, Prediction, ML, Detection, Healthcare

I. INTRODUCTION

When specific regions of the brain experience blockages and blood flow become obstructed, a stroke develops [1]. Within minutes, nerve cells perish due to the blood's inability to supply them with oxygen, glucose, and nutrients for proper functioning. To mitigate the risks of a brain stroke, early detection is essential. Once the risk of a stroke has been identified, it is crucial to anticipate potential outcomes and consequences in the event of a stroke occurrence. Stroke ranks as the second most prevalent cause of morbidity and mortality globally [2]. It is a significant global health issue, with various risk factors and outcomes, emphasising the

importance of accurate risk prediction and outcome assessment to guide clinical decision-making processes and significantly enhance overall patient outcomes [1][2]. ML techniques have appeared as valuable tools for analysing complex data and predicting stroke risk and post-stroke outcomes. Additionally, the escalating expenses associated with stroke-related hospitalisations underscore the urgency of deploying technological solutions capable of facilitating clinical diagnosis, treatment, prognostication of clinical events, formulation of effective treatment regimens, and implementation of rehabilitation programs, among other critical functions [1].

The domain of ML techniques for the prediction and detection of stroke risks as well as post-stroke outcome has received much attention. Significant advancement have been made in improving the precision and reliability of identifying individuals susceptible to stroke and their post stroke outcomes. While prior research has largely focused on either detecting stroke risks or predicting post-stroke outcomes, this study seeks to integrate these two aspects to provide a holistic understanding of stroke management. By synthesising findings from a wide range of studies, this comprehensive review aims to answer the following questions and achieve the following review objectives.

A. Review Objectives

- Provide a comprehensive overview of existing literature on ML applications for stroke risk detection and post-stroke outcome prediction.
- Evaluate the performance and reliability of various ML algorithms to detect stroke risk factors and predict post-stroke outcomes.
- Discuss the implications of ML-based stroke management approaches, including their potential impact on clinical decision-making, public health strategies, and future research directions.

B. Review Questions

- What ML techniques and methods have been explored to detect stroke risk factors and predict post-stroke outcomes?
- How do various ML algorithms perform in identifying individuals at risk of stroke and predicting their outcomes post-stroke?

- c) What are the challenges and implications of implementing ML-based approaches in stroke risk assessments and post-stroke outcome prediction?

The paper is organised to encompass an introduction, a review of related works focusing on existing reviews in this field, the methodology utilised in this study, which includes the study design, search strategy, and selection criteria for the reviewed papers. Subsequently, it delves into the results and discussion, explores prospects for future research, and concludes with key findings.

II. RELATED WORKS

Numerous studies in the literature have explored diverse facets of stroke prediction, detection, and subsequent outcomes. This section recognises the valuable contributions made by fellow researchers in this field. It encompasses systematic literature reviews, comprehensive analyses, and other relevant review works on ML methodologies for detecting and predicting stroke risks and post-stroke outcomes.

Li et al. [2] conducted a systematic review, aiming to evaluate ML models for predicting motor function post-stroke in their study. They highlighted ML's capacity to process non-linear data and improve prediction accuracy. Employing the Prediction model Risk Of Bias ASsessment Tool (PROBAST) for quality assessment alongside a meticulous literature search, the review offers profound insights into prevailing methodologies, techniques, and predictive accuracies achieved by various ML approaches. While the paper provides a comprehensive overview, further clarification on specific ML algorithms and a detailed explanation of PROBAST criteria would enhance its clarity and credibility.

Campagnini et al. [3] conducted a systematic review to explore the utility of ML algorithms in predicting motor functional recovery among post-stroke patients undergoing rehabilitation. Through a thorough search across five electronic databases using the PICO (Patient/Population, Intervention, Comparison and Outcomes) format, the study included 19 primary studies. These studies mainly leaned on demographic features and stroke evaluations conducted during clinical examinations as predictive factors. Among the methodologies employed, linear and logistic regressions emerged as the predominant choices. Despite methodological limitations, such as small sample sizes and limited external validation approaches, the review provides insights into applying ML algorithms in rehabilitation medicine.

Schwartz et al. [4] systematically examined the performance of ML models in predicting mortality after stroke, aiming to identify key factors associated with mortality. Through a comprehensive search spanning MEDLINE/PubMed and Web of Science databases, 28 pertinent studies were included, revealing favourable AUC ranges (0.67–0.98) for mortality prediction by ML models. Notably, age, high Body Mass index (BMI), and elevated National Institutes of Health Stroke Scale (NIHSS) scores surfaced as significant mortality predictors. However, the study underscores prevalent bias risks and concerns regarding the applicability of findings across the reviewed literature, stressing the imperative for prospective, multicentre studies with standardised reporting to fortify the robustness of ML algorithms in stroke prognostication.

In another study by Lucas et al. [5], the authors critically reviewed stroke risk prediction systems utilising ML techniques. They emphasise ML's potential as a

complementary tool for neurologists and stress the importance of early stroke detection by identifying modifiable and nonmodifiable risk factors. Despite developing various ML-based prediction systems, the authors note that data pre-processing techniques need to be improved in many studies. However, they observe a trend towards integrating graphical user interfaces (GUIs) to enhance accessibility for healthcare professionals when presenting prediction results. The review calls for improved implementation of ML algorithms through standardised guidelines and reporting standards to enhance prediction reliability and acceptance. Additionally, the study highlights the importance of factor assessment in addressing transparency issues in ML models, particularly in demonstrating the contribution of risk factors to predictions.

III. METHODOLOGY

A. Study Design

This study's primary aim is to comprehensively review the contribution of ML applications in predicting and detecting stroke and post-stroke outcomes. This approach is systematic as it critically evaluates and reviews the relevant studies, synthesise their findings, and presents a comprehensive result and discussion section, offering valuable insights from multiple sources. Beyond examining current applications and reporting on the frequently used ML algorithms, techniques, and datasets across the studies, this study delves into prospects and unanswered research queries and identifies opportunities for advancing stroke research. This comprehensive analysis aims to guide the key areas that future research should prioritise and focus on.

B. Search Strategy

A comprehensive search for available papers and publications related to the usage of ML to predict the risk of stroke and post-stroke outcomes was conducted on the IEEE Xplore Digital Library database, PubMed, ScienceDirect, SpringerLink and Web of Science database. The search strings incorporating keywords related to stroke prediction, detection, machine learning, and post-stroke outcomes were utilised in three sequential searches: a general search, a prediction-focused search, and a detection-focused search. Relevant publications underwent manual review to extract pertinent evidence for analysis. Titles and abstracts of potentially relevant studies were then scrutinised, and articles were selected according to their alignment with the search criteria. The search process involved utilising a range of keywords and logical operators (such as AND and OR) to explore relevant publications spanning 9 years. Notably, no relevant studies were identified or available in digital databases prior to 2016. The search encompassed both titles and abstracts of the papers to ensure a thorough exploration of relevant literature.

1) General Search String:

(Machine Learning) AND ((Detect* OR Predict*) AND (Stroke Risk OR Post-Stroke Outcomes))

2) Prediction Focused Search String:

(Machine Learning) AND (Predict*) AND (Stroke Risk OR Post-Stroke Outcomes)

3) Detection Focused Search String:

(Machine Learning) AND (Detect*) AND (Stroke Risk OR Post-Stroke Outcomes)

C. Selection of Studies

Following the retrieval of articles from the digital databases, an Excel file was created to systematically record the papers included in the study. This file captured essential details such as the paper title, publication year, utilised dataset, explored ML methods or techniques, methodology used, tools and techniques employed, and model evaluation accuracy and computational efficacy metrics derived from all fully reviewed papers. Additionally, the Mendeley reference manager software was employed to establish a comprehensive repository of references, eliminate duplicates, and assist in managing citations for the papers included in this research.

D. Inclusion Criteria

Table 1 outlines the inclusion criteria established for this study. Only manuscripts written in English and published as conference papers or journal articles between the years 2016 and 2024 were considered. The emphasis was placed on works that discuss the applications of ML methods and techniques for the detection and prediction of stroke risk, as well as post-stroke outcomes.

TABLE 1: INCLUSION CRITERIA

Criteria	Definition
Manuscript Language	English
Accepted Paper Types	Conference papers, Journal publications
Time Period/ years	2016-2024
Emphasis	Applications of ML methods and/or techniques for detection and/or prediction of stroke risk and/or post stroke outcomes

E. Exclusion Criteria

This study excluded papers meeting specific criteria, such as review papers and those lacking full-text availability. Additionally, this comprehensive review filtered out theses, books, editorials, essays, chapters, governmental guidelines, non-English publications, non-peer-reviewed materials, and studies not aligning with the criteria outlined in Table 1. While review papers were not incorporated into the analysis, they were separately reviewed for relevant content. Furthermore, papers discussing stroke detection or prediction without ML methods were excluded.

IV. RESULTS AND DISCUSSION

A. ML trends in stroke risk detection and post-stroke outcome prediction

Figure 1 depicts the evolving patterns in publications regarding ML applications for predicting and detecting stroke risks and post-stroke outcomes. The figure distinctly reveals a notable surge in research publications, especially in recent years. There has been a considerable uptick in paper publications, notably peaking in 2023. These findings highlight a rising interest, emphasising the growing momentum and acknowledgement of the potential of ML applications in detecting and predicting stroke risks and post-stroke outcomes.

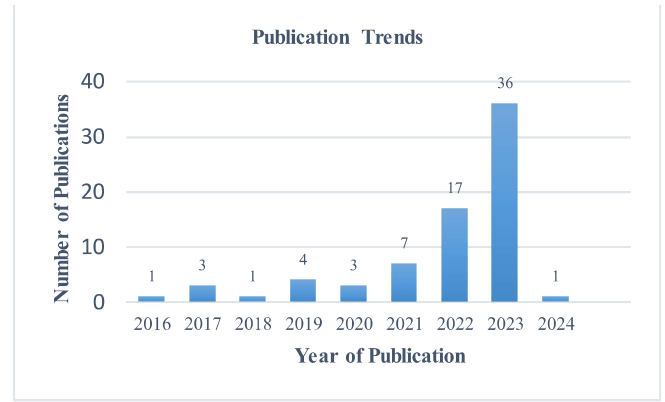


Figure 1: Publication Trends

B. ML techniques and methods stroke risk detection and post-stroke outcome prediction.

This review reveals that thirty-nine (39) ML techniques have been explored in this field. Within this set of techniques, Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, Naïve Bayes and k-Nearest Neighbors (k-NN) have surfaced as the foremost commonly utilised techniques, as illustrated in Figure 2. These ML algorithms and techniques have been preferred because they handle diverse datasets, offer robust predictive capabilities, effectively discern patterns, and provide valuable and interpretable insights. These qualities make them indispensable tools in advancing our understanding of stroke-related phenomena.

The extensive exploration of Random Forest, SVM, Logistic Regression, Decision Tree, Naïve Bayes, and k-NN underscores their significance in the domain of ML models concerning predicting and detecting stroke risks and post-stroke outcomes. With its robustness and capacity to handle intricate datasets, Random Forest excels in identifying patterns associated with stroke risks and outcomes [6][7]. SVM's versatility in handling linear and nonlinear data and its ability to delineate complex decision boundaries make it invaluable for stroke risk prediction tasks [6], [8]. Despite its simplicity, Logistic Regression remains a cornerstone in medical research, offering interpretability and efficiency in estimating probabilities related to stroke-related data [9][10].

Decision Trees provide transparent frameworks for modelling complex relationships within datasets, identifying risk factors, and predicting outcomes [11]. Naïve Bayes classifiers, known for their simplicity and efficiency in handling high-dimensional data, offer probabilistic approaches ideal for stroke risk assessment and outcome prediction [12]. k-NN's simplicity and effectiveness in classification tasks, leveraging similarities between data points, make it useful for discerning relevant patterns in stroke-related datasets [6], [13]. Collectively, these models contribute to advancing stroke research by providing insights, aiding risk assessment, and facilitating early detection and intervention strategies, reflecting their importance in addressing the complex challenges associated with stroke prediction and post-stroke outcome detection.

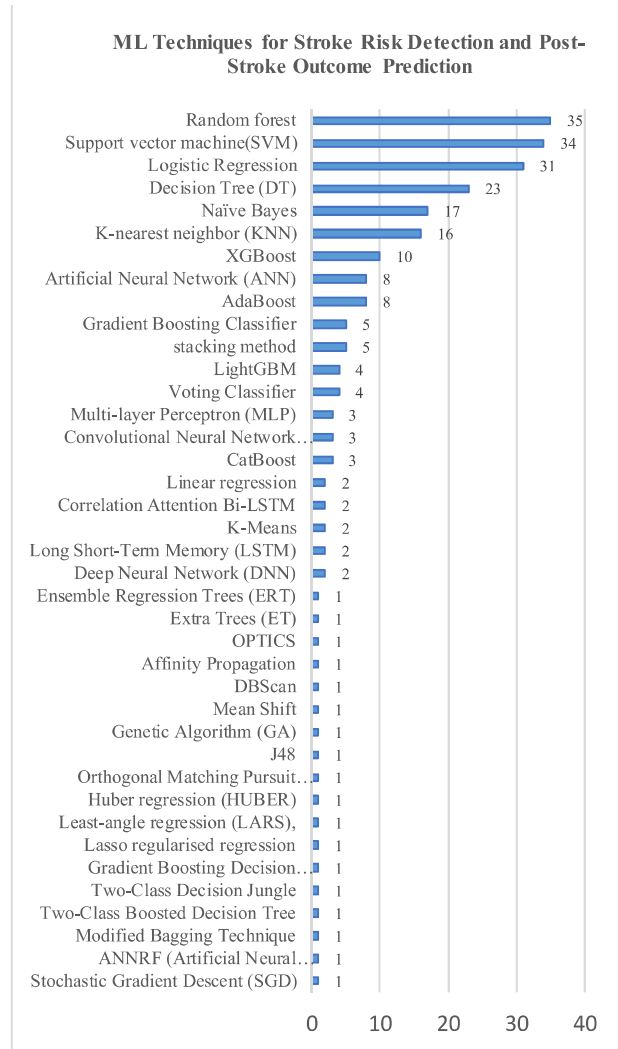


Figure 2: ML Techniques Explored

Table 2 illustrates the prevalence of various ML algorithms employed in the studies reviewed, along with their corresponding reference paper IDs and frequency of occurrence. Random forest emerges as the most frequently utilised method, cited across 35 papers, followed closely by SVM with 34 mentions. Logistic Regression and Decision Tree methods are also prominent, appearing in 31 and 23 papers respectively. Other techniques such as Naïve Bayes, k-NN, and XGBoost demonstrate notable usage patterns, underlining the diverse array of methodologies employed in stroke risk assessment and post-stroke outcome prediction research. Moreover, emerging methods such as Convolutional Neural Network (CNN), Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) are gaining traction, signaling a growing interest in leveraging advanced deep learning architectures for predictive modeling in stroke-related studies as shown in Figure 2 as well.

ML method/Technique	Paper ID	Frequency
Random forest	[6][7][14][9][15][13][16][17][18][10][12][19][20][21][22][23][24][25][26][27][28][29][30][31][32]	35
Support vector machine(SVM)	[6][11][7][14][9][15][13][16][8][17][18][19][20][33][23][23][24][25][26][27][34][28][35][36][37][31][32]	34
Logistic Regression	[6][11][9][15][13][8][17][18][10][38][39][19][20][21][24][25][26][27][34][29][37][40][32]	31
Decision Tree (DT)	[6][11][7][14][9][13][41][8][17][18][10][38][21][24][27][34][35][37][30][32]	23
Naïve Bayes	[6][13][16][8][17][18][12][38][24][27][28][37]	17
K-nearest neighbor (KNN)	[6][11][7][9][13][18][12][24][27][34][28][36][29][37]	16
XGBoost	[11][7][9][15][42][18][38][43][19][24][25][27][44][35][36][31]	10
AdaBoost	[11][38][43][24][27][34][36]	8
Artificial Neural Network (ANN)	[10][20][25][34][36]	8
stacking method	[6][45][9][46][13][39][21][44]	5
Gradient Boosting Classifier	[38]	5
Voting Classifier	[27][36]	4
LightGBM	[11][15][35]	4
CatBoost		3
Convolutional Neural Network (CNN)	[6][9][47][48]	3
Long Short-Term Memory (LSTM)	[47][49]	2
K-Means	[23][24][28]	2
Correlation Attention Bi-LSTM	[50][48]	2
Linear regression	[37]	2
Stochastic Gradient Descent (SGD)	[36]	1
ANNRF (Artificial Neural Network - Random Forest)	[51]	1
Gradient Boosting	[34]	1

Decision Trees (GBDT)		
Lasso regularised regression	[52]	1
Least-angle regression (LARS),	[52]	1
Huber regression (HUBER)	[52]	1
Orthogonal Matching Pursuit (OMP)	[52]	1
J48	[12]	1
Genetic Algorithm (GA)	[47]	1
Mean Shift	[23]	1
DBScan	[23]	1
Affinity Propagation	[23]	1
OPTICS	[23]	1
Extra Trees (ET)		1
Ensemble Regression Trees (ERT)	[16]	1

C. Dataset utilised

When constructing and building ML models for stroke risk detection and post-stroke outcome prediction, selecting a dataset or database is crucial as it forms the foundation for model training, evaluation, and validation. Scholars have utilised a range of datasets in their studies to develop ML models customised for predicting and detecting stroke risks and post-stroke outcomes. Notably, the Kaggle database, real dataset, Framingham and National Health Insurance Research Database (NHIRD) have surfaced as the most commonly used, as illustrated in Figure 3.

Kaggle offers diverse datasets across wide range domains and real datasets provides valuable insights into actual phenomena and behaviors observed in clinical practice or real-world settings. Furthermore, the Framingham Heart Study dataset and the NHIRD have garnered attention for their comprehensive coverage of health-related variables and long-term follow-up data. These datasets databases are crucial in advancing stroke prediction and management research, enabling researchers to develop and validate ML models that can assist in early detection, risk assessment, and personalised intervention strategies for stroke patients.

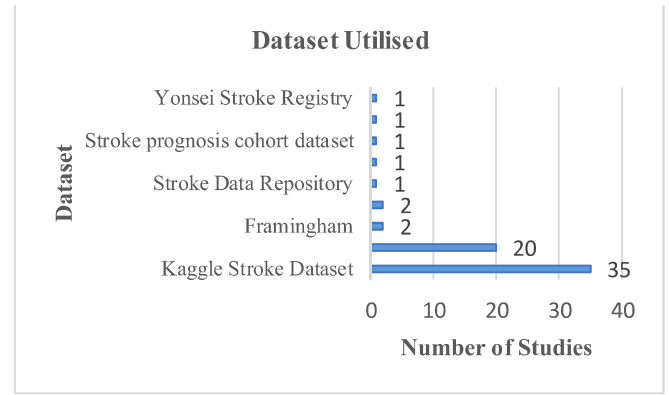


Figure 3: Dataset Used

D. Tools and Techniques Used

A total of 10 distinct tools and techniques were explored across the papers included in this study. These encompass Python (utilising Scikit-Learn, GridSearchCV, Pandas, and Matplotlib), PyCharm, MATLAB, Anaconda, Flask, Apache Spark, IBM SPSS, WEKA, Visual Studio Code (VSCode), and R. Python has emerged as the dominant choice and the most extensively utilised technique for constructing ML models to predict stroke risks and post-stroke outcomes. Within Python libraries, Scikit-Learn has garnered considerable attention, establishing itself as the premier choice among researchers. Scikit-Learn offers user-friendly and effective tools for data mining and analysis, utilising foundational libraries like NumPy, SciPy, and Matplotlib. Moreover, it offers an extensive selection of effective ML and statistical modeling resources, including diverse algorithms, evaluation metrics, preprocessing methods as well as robust support for tasks like data transformation, feature selection, model selection, and hyperparameter tuning.

E. ML Pros and Cons in Stroke Detection and Prediction

In assessing the pros and cons of ML in the detection and prediction of stroke risks and post-stroke outcomes, several key themes emerge from the literature. ML techniques offer significant advantages in their ability to process large amount of complex datasets, identify hidden patterns, handle multidimensional data effectively and produce highly accurate predictive models [3]. These models enable early detection of stroke risk factors, prediction of stroke occurrence, and prognosis of post-stroke outcomes, thereby facilitating timely interventions and personalised patient care [53] [37]. Additionally, ML models can incorporate a wide range of input features, such as demographic, clinical, and imaging data, allowing for comprehensive risk assessment and outcome prediction.

Furthermore, ML algorithms can provide insights into the underlying mechanisms of stroke pathology and inform clinical decision-making by identifying relevant biomarkers and risk factors [54]. However, challenges such as class imbalance, data heterogeneity, interpretability of models, and generalisability to diverse populations remain significant limitations in the application of ML for stroke prediction and outcome assessment [6]. Moreover, the black-box nature of some ML algorithms poses challenges in understanding the rationale behind model predictions, potentially hindering their clinical adoption and trust among healthcare providers [55]. Despite these limitations, ongoing research efforts focused on model explainability, interpretability, and validation in real-world clinical settings hold promise for enhancing the utility

and reliability of ML-based approaches in stroke risk prediction and outcome assessment [56].

F. Performance Analysis of ML Techniques

Evaluation metrics such as accuracy, recall, precision, sensitivity, F1-score, specificity, area under the ROC curve (AUC), confusion matrix, Matthews Correlation Coefficient (MCC), mean absolute error (MAE), root mean square error (RMSE), and log loss were utilised to assess the performance and reliability of various machine learning algorithms for stroke risk detection and post-stroke outcome prediction. Several algorithms demonstrated impressive performance across these metrics. For instance, the Stacking Classifier exhibited outstanding results with an accuracy of 99.89%, precision of 99.79%, recall of 100%, and an F1-score of 99.89% [6]. Similarly, the Convolutional Neural Network (CNN) achieved notable results with an accuracy of 96.55%, precision of 93.88%, recall of 99.59%, and an F1-score of 96.65% [6][13][47]. Notably, ensemble methods and hyperparameter tuning demonstrated substantial improvements in accuracy, with some models achieving up to 99% accuracy [7][45][51][9].

Additionally, techniques such as feature selection, resampling, normalisation, filtering and preprocessing played pivotal roles in enhancing model performance and addressing challenges like class imbalance [6][45][7]. Execution time and prediction time are practical considerations, especially for real-time applications or large-scale deployments [45]. While some algorithms like Decision Tree and Random Forest exhibited faster execution times, others such as SVM and Neural Network models may require more computational resources but achieve higher accuracy [45]. These performance results and findings underscored the potential effectiveness of ML in enabling early detection, accurate prediction, and personalised treatment approaches for stroke patients.

G. Challenges and Implications for ML in Clinical Settings

When considering the integration of ML in clinical settings, several challenges and implications emerge that warrant careful consideration. Foremost among these is the complexity of healthcare data, which often varies widely in structure and quality across different sources [55]. Standardising and ensuring the integrity of this data is crucial for the development, evaluation and validation of ML models. Additionally, the interpretability of ML algorithms remains a key concern, especially in healthcare where transparency is essential for gaining trust from clinicians and patients alike [57].

Addressing ethical and regulatory considerations, like patient privacy and algorithmic bias, is also paramount to warrant equitable and liable utilisation of ML in healthcare [55]. Ensuring compliance and adherence with regulations such as Health Insurance Portability and Accountability Act (HIPAA) is crucial for protecting patient confidentiality. Moreover, integrating ML seamlessly into clinical workflows requires careful consideration of factors like clinician training, system interoperability, and long-term sustainability.

V. PROSPECTS FOR FUTURE RESEARCH

The domain of ML techniques for the prediction and detection of stroke risks, as well as post-stroke outcomes, has

received much attention. Significant advancements have been made in improving the precision and reliability of identifying individuals susceptible to stroke and their post-stroke outcomes. However, numerous promising directions for future research warrant exploration to improve effectiveness within stroke. Figure 4 outlines potential future research directions for ML techniques in detecting stroke risks and predicting post-stroke outcomes.

The next area of research that needs to be explored is transfer learning and domain adaptation, dynamic modelling and temporal analysis to capture the evolving nature of stroke risk factors. Additionally, there is a pressing need to rectify data imbalances and biases fostered by developing hybrid models that integrate diverse data sources. Embracing multi-modal approaches and data fusion techniques could enhance predictive accuracy by leveraging complementary information streams. Moreover, it is essential to thoroughly ponder the ethical and societal implications associated with employing ML for stroke risk assessment and post-stroke outcome prediction, as depicted in Figure 4.

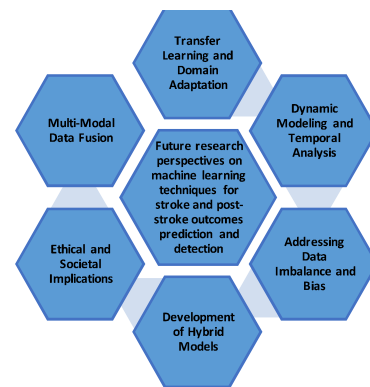


Figure 4: Future Trends for ML Applications in Stroke Prediction and Detection

VI. CONCLUSION

In summary, this comprehensive review provides insights into the contemporary landscape of ML applications concerning stroke prediction and the assessment of post-stroke outcomes. Analysing forty-six research papers from 2016 to 2024 identifies prominent ML techniques such as Random Forest, SVM, and Logistic Regression, emphasising Python's dominance as the preferred platform. Looking forward, the review suggests exploring avenues like transfer learning, dynamic modelling, and addressing data biases to enhance the effectiveness of ML in stroke research. By embracing these future directions, the field can advance towards more accurate predictions, informed clinical decisions, and improved patient care outcomes.

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